

Federal Agency: Army

Program Office: Army Contracting Command-Red River Army Depot (ACC-RRAD)

AoI Title: Road Wheel Drum Tester, Bldg 493

CSO AoI Number: 001

AoI Release Date: 8 Oct 2024

Responses: See W911RQ-24-S-C0001 for Procedure Details

General Information

Introduction

This announcement is for the design, manufacture, and installation of a Road Wheel Drum Tester Area of Interest (AoI) under the ACC-RRAD Commercial Solutions Opening W911RQ-24-S-C0001. Interested parties are highly encouraged to visit the Government Point of Entry (www.sam.gov) to review the ACC-RRAD CSO special posting in its entirety to obtain more information on the process for submission and evaluation of the CSO AoI Responses.

This CSO AoI seeks to solicit innovative solutions that meet the objectives detailed in Section 1.2 and 1.3. The term innovative is defined as any technology, process, or method, including research and development that is new as of the date of proposal submission; or any new application of an existing technology, process or method as of the proposal date.

ACC-RRAD is not obligated to extend a request for proposal or make an award as a result of this CSO AoI; all awards are subject to the availability of funds and successful negotiations.

This CSO AoI is not an authorization to begin performance, and in no way obligates the Government for any costs incurred by the Offeror associated with developing a Response. Prior to commencement of any activities associated with performance under this CSO AoI, the Government will issue a written directive or contractual document signed by the Contracting Officer with appropriate consideration established. ACC-RRAD reserves the right to modify the requirements of this CSO AoI.

The authority to issue this CSO AoI is derived from ACC-RRAD Commercial Solutions Opening W911RQ-24-S-C0001, dated 16 October 2022. The CSO authority is Section 803 of the National Defense Authorization Act for FY2022, Public Law 117-807, and Class Deviation 2022-O0007.

1.2 Aol Background

Red River Army Depot (RRAD) Rubber Products Division has a need for the design, manufacture, and installation of a Road Wheel Drum Tester to conduct tests on military vehicle road wheels (see Fig 1 Below).



Figure 1 - Road Wheel Sets

1.3 Improved Road Wheel Drum Tester

RRAD is interested in a commercial solutions offerings for the design, manufacture, and installation of a Road Wheel Drum Tester that will improve upon the current process. We are looking for advanced manufacturing capabilities to improve upon the current process of quality testing of road wheels. These assets are manually inspected and require other quality test to determine if the production processes produce assets within quality parameters outlined in MIL-DTL-3100H, and DMWR 9-2530-200-1 (Current revisions). RRAD is looking to improve this process to enhance our safety, capabilities and capacity and to reduce maintenance / operating costs involved with the current process.

At a minimum RRAD is requesting a new Road Wheel Drum Tester to complete the required tests as described below (Reference: MIL-DTL-3100H).

- Road Wheel, Track Support Rollers, and Idler Endurance Tests (MIL-DTL-3100)
- Road Wheel Blowout Test (MIL-DTL-3100)

The following system requirements must also be met:

- The drum must have a diameter of 67.23" +/- 0.5" (as per MIL-DTL-3100H)
- The width of the drum rim must provide full contact with the road wheel rubber tread.
- The max run test is 48 hours long, run under a road wheel loading of 2,344lbs at minimum, and 6,210lbs at maximum.
- The 6 hour test has a minimum wheel loading of 1,242lbs and a maximum of 3,276lbs.
- The destructive test is up to three hours long, run under a road wheel pressure of 5,500lbs.
- Wheel loading pressure must be adjustable from the operator work station.
- The drum tester must be able to run two road wheels simultaneously at different weight loadings.
- The wheel speed range is 10mph up to 35mph.
- The new system must have safety features like multiple emergency stops, and a way to determine when the road wheel has failed and shut the test down automatically. Currently, we use a wire across the road wheel to determine when the rubber has failed and automatically activate the break and end the test.
- The new system must be able to provide 3 temperature sensor readings on the road wheel while it is being run: Inner metal surface of wheel, outer rubber face of the wheel (the crown), and the side profile of the wheel rubber.
- The new system must include provisions for safely lifting and loading/unloading road wheels from the test fixtures.
- At some point in the future RRAD may be required to load test road wheels at pressures up to 12,000lbs at maximum, and all new test stands must be able to handle this.

The system shall process all test conditions described into a central computer capable of receiving the data and processing reports with exports to Excel. The new software that runs the new equipment must also be able to produce trend graphs, pinpoint test results, save data digitally, and also print out hardcopies. The software must be able to log test runtime, high/low 3 temp sensor data (3 road wheel temp sensors), room temperature, test date, test time, start time, finish time, wheel load (lbs), and a few user input fields such as test type, wheel size, rubber mfg, and other user test input information to be defined at a later date.

Aol Constraints

Any selected company/companies will be required to be registered in the System for Award Maintenance (sam.gov).

Submission Requirements

A written solution brief must be sent via e-mail submission by 5:00 P.M. (CDT) on 7 Nov 2024, with subject line stating "Remove & Replace Road Wheel Drum Tester System" to adam.s.ragland.civ@army.mil and usarmy.rrad.acc.mbx.cso@army.mil.

Solution briefs received after the specified due date and time will be discarded and not be evaluated by the Government.

Aol Open Period – This AoI shall close at 5:00 P.M (CDT); 7 Nov 2024.

Point of contacts:

Aol questions can be sent to:

Contracting Officer:

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CSO AoI POCs:

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